

Linear Algebra I: HW3 Partial Solution

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1. FTFFFTFFFTTTT

14. $\beta = \{(1, 0, 1, 0, 0), (1, 0, 0, 1, 0), (0, 1, 0, 0, 0), (0, 0, 0, 0, 1)\}$ is an F -basis for W_1 , $\dim_F W_1 = 4$

$\gamma = \{(1, 0, 0, 0, -1), (0, 1, 1, 1, 0)\}$ is an F -basis for W_2 , $\dim_F W_2 = 2$.

24. $\beta = \{f, f', f'', \dots, f^{(n)}\}$ is linear independent since each $f^{(i)}$ has distinct degree. It is an $\mathbb{R}$ -basis since $\dim_{\mathbb{R}} P_n(\mathbb{R}) = n + 1 = |\beta|$.

29. (a). Let γ be a basis for $W_1 \cap W_2$, which is a finite dimensional vector subspace of V , and extend γ to a basis for W_1 (resp. W_2), call it β_1 (resp. β_2). Write

$$\gamma = \{v_1, \dots, v_n\}$$

$$\beta_1 = \{v_1, \dots, v_n, s_1, \dots, s_m\}$$

$$\beta_2 = \{v_1, \dots, v_n, u_1, \dots, u_k\}$$

We claim that $\eta = \{v_1, \dots, v_n, s_1, \dots, s_m, u_1, \dots, u_k\}$ is a basis for $W_1 + W_2$. Since $\text{span}(\eta) = \text{span}(\beta_1 \cup \beta_2) = \text{span}(\beta_1) + \text{span}(\beta_2) = W_1 + W_2$, η is a generating set of $W_1 + W_2$ (this also proves that $W_1 + W_2$ is finite dimensional). It suffices to show that η is linearly independent. Suppose that there is a relation

$$a_1 v_1 + \dots + a_n v_n + b_1 s_1 + \dots + b_m s_m + c_1 u_1 + \dots + c_k u_k = 0.$$

for some $a_1, \dots, a_n, b_1, \dots, b_m, c_1, \dots, c_k \in F$. By moving u terms to the other side, we have

$$a_1 v_1 + \dots + a_n v_n + b_1 s_1 + \dots + b_m s_m = -(c_1 u_1 + \dots + c_k u_k).$$

Therefore, $c_1 u_1 + \dots + c_k u_k \in W_1$, and $c_1 u_1 + \dots + c_k u_k \in W_2$, so $c_1 u_1 + \dots + c_k u_k \in W_1 \cap W_2$ and can be written as

$$c_1 u_1 + \dots + c_k u_k = d_1 v_1 + \dots + d_n v_n$$

for some $d_1, \dots, d_n \in F$. That is, we have a relation

$$-(d_1 v_1 + \dots + d_n v_n) + c_1 u_1 + \dots + c_k u_k = 0.$$

Since β_2 is linearly independent, we have $d_1 = \dots = d_n = c_1 = \dots = c_k = 0$, which also implies $a_1 = \dots = a_n = b_1 = \dots = b_m = 0$ since β_1 is linearly independent. Similarly, we can deduce that $a_1 = \dots = a_n = c_1 = \dots = c_k = 0$. Therefore, η is linear independent and

$$\begin{aligned} \dim(W_1 + W_2) &= |\eta| = n + m + k \\ &= |\beta_1| + |\beta_2| - |\gamma| \\ &= \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2). \end{aligned}$$

(b). If $V = W_1 + W_2$, then

$$\begin{aligned} V = W_1 \oplus W_2 &\iff W_1 \cap W_2 = \{0\} \\ &\iff \dim(W_1 \cap W_2) = 0 \\ &\iff \dim V = \dim W_1 + \dim W_2. \end{aligned}$$

30. $\beta_1 = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$ is a basis for W_1 .

$\beta_2 = \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is a basis for W_2 .

$W_1 \cap W_2 = \left\{ \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} \mid a \in F \right\}$, and $\gamma = \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$ is a basis for $W_1 \cap W_2$.

Combining all the results, we have

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2) = 3 + 2 - 1 = 4.$$

(This implies that $W_1 + W_2 = V$.)

33. (a). If $\beta_1 \cap \beta_2$ is not empty, then there exists a nonzero element $u \in \beta_1 \cap \beta_2$, but then $u \in W_1 \cap W_2 = \{0\}$, which is absurd. Therefore, $\beta_1 \cap \beta_2 = \emptyset$.

Since $\beta_1 \cup \beta_2$ spans V and

$$\dim V = \dim W_1 + \dim W_2 = |\beta_1| + |\beta_2| = |\beta_1 \cup \beta_2|$$

$\beta_1 \cup \beta_2$ is a basis for V .

(b). Since we have

$$V = \text{span}(\beta_1 \cup \beta_2) = \text{span}(\beta_1) + \text{span}(\beta_2) = W_1 + W_2.$$

It suffices to show that $W_1 \cap W_2 = \{0\}$, suppose not, then there is a nonzero element $u \in W_1 \cap W_2$, and u can be written as a nontrivial linear combination of β_1, β_2 , which leads to a nontrivial relation among $\beta_1 \cup \beta_2$, and the contradiction occurs since $\beta_1 \cup \beta_2$ is assumed to be linear independent.

34. (a). Let γ be a basis for W_1 and extend γ to be a basis β for V . Put $\eta = \beta \setminus \gamma$, then $W_2 = \text{span}(\eta)$ is a subspace of V . Furthermore, $V = W_1 \oplus W_2$ from Exercise 33(b).

Remark. In this case, we say that V is **semisimple**.

(b). $W_2 = \{(0, y) \mid y \in \mathbb{R}\}$, $W'_2 = \{(y, y) \mid y \in \mathbb{R}\}$.